ScienceDirect

Energy Storage Materials
Volume 81, September 2025, 104469

Dynamic interface regulation in solid-state lithium-metal batteries by in situ polymerized highly elastic ultrathin layers

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https://doi.org/10.1016/j.ensm.2025.104469 ↗

Abstract

Poly(vinylidene fluoride) (PVDF)-based solid-stated electrolytes (SSEs) hold great promise for high-energy-density solid-state lithium metal batteries (SSLMBs) due to their relatively high ionic conductivity and excellent flexibility. However, their practical performance is hindered by instability at the Li/PVDF interface, where the porous PVDF induces uneven ion deposition and residual N,N-dimethylformamide (DMF) solvent triggers severe side reactions with Li. Here, we demonstrate that a highly elastic layer, introduced via in situ polymerization of 1,3-dioxolane (P-DOL) containing thermoplastic polyurethane (TPU), effectively stabilizes and dynamically regulates the Li/PVDF interface. The incorporation of TPU enhances the elastic modulus of the cured layer, enabling local stress redistribution and dynamic retention of intimate interfacial contact at the Li/PVDF interface during cycling, which in turn promotes dense and flat Li deposition morphology to prolong the cycle life. Moreover, the interlayer prevents side reactions between DMF residues and Li anode, preserving an electrochemically stable contact. Consequently, the Li|PVDF|$LiNi_{0.8}Co_{0.1}Mn_{0.1}O_2$ cell with TPU/P-DOL interlayer delivers a capacity of 136 mA h g^{-1} over 800 cycles at 1 C. Our findings demonstrate a scalable interfacial engineering strategy that addresses the key bottleneck of PVDF-based SSEs and significantly advances the performance of SSLMBs.

Previous

Next

Keywords

Lithium metal batteries; Li/PVDF interface; In situ polymerization; Thermoplastic polyurethanes; Elastic interfacial layers

1. Introduction

Lithium (Li) metal batteries (LMBs) have garnered considerable attention as prospective alternatives to traditional Li-ion batteries due to their higher energy density facilitated by the substitution of graphite-based anodes with Li metal, which offers a superior specific capacity of 3860 mA h g^{-1} and the lowest standard reduction potential of all metals of −3.04 V vs. standard hydrogen electrode [1,2]. Such a surge in energy density is urgently required to satisfy the needs of various high-energy-density storage applications, such as boosting the electric vehicle range and facilitating a secure grid integration of non-dispatchable renewable energy sources. However, the first attempts to commercialize LMBs by companies such as Moly Energy in the 1980s failed due to serious safety issues associated with the formation of Li dendrites on the anode side, which can penetrate the separator and lead to internal short circuits, thermal runaway, and even fires [3].

One promising strategy to address these challenges involves the advancement of the solid-state electrolyte (SSE) as an alternative to conventional flammable liquid electrolytes, aiming at attenuating dendrite formation and fortifying the safety of LMBs [4]. Among these, inorganic SSEs exhibit relatively high ionic conductivity, yet their susceptibility to brittleness (especially for oxides), vulnerability to hydration, limited electrochemical window (for sulfides and halides) and inadequate interparticle as well as electrolyte/electrode interfacial contact hinder their widespread scalability and applicability [5,6]. In contrast, organic SSEs such as poly(vinylidene difluoride) (PVDF) polymers combine similar ionic conductivities with superior performance in terms of wider electrochemical window, excellent interfacial wettability and mechanical flexibility [7]. Uniquely, due to the limited ability of the PVDF molecular chains to dissociate the lithium salts and its strong interactions with trace residuals of N,N-dimethylformamide (DMF) solvent, PVDF-based electrolytes primarily facilitate Li^+ conduction through the transportation of solvated $[Li(DMF)_x]^+$ complexes by the segmental motion of polymer chains. Notably, different from the typical gel electrolytes that contain >50 wt % free solvent, this PVDF SSE contains almost no free DMF residual solvent and most of the DMF is bonded to Li^+[8].

Nevertheless, several challenging issues associated with PVDF-based electrolytes cannot be overlooked (Fig. 1b). Firstly, PVDF-based electrolytes exhibit a porous rather than fully dense structure due to the phase separation between polymer and solvent during drying, which causes a heterogeneous Li^+ flux at the interface. As a result, both Li^+ de/insertion at the cathode/electrolyte interface and Li metal deposition/stripping at the anode side tend to be non-uniform, leading to a rapid capacity degradation and Li dendrite growth. In addition, free trace amounts of DMF solvent at the interfaces trigger severe side reactions with the Li metal anode resulting in a thicker interfacial layer [9,10]. Likewise free DMF undergoes oxidative decomposition when matched with high-voltage cathode materials such as $LiNi_{0.8}Co_{0.1}Mn_{0.1}O_2$ (NCM811), further increasing the interfacial impedance [11]. Addressing the aforementioned issues is imperative to further propel the practical applicability of the PVDF-based SSE in solid-state batteries.

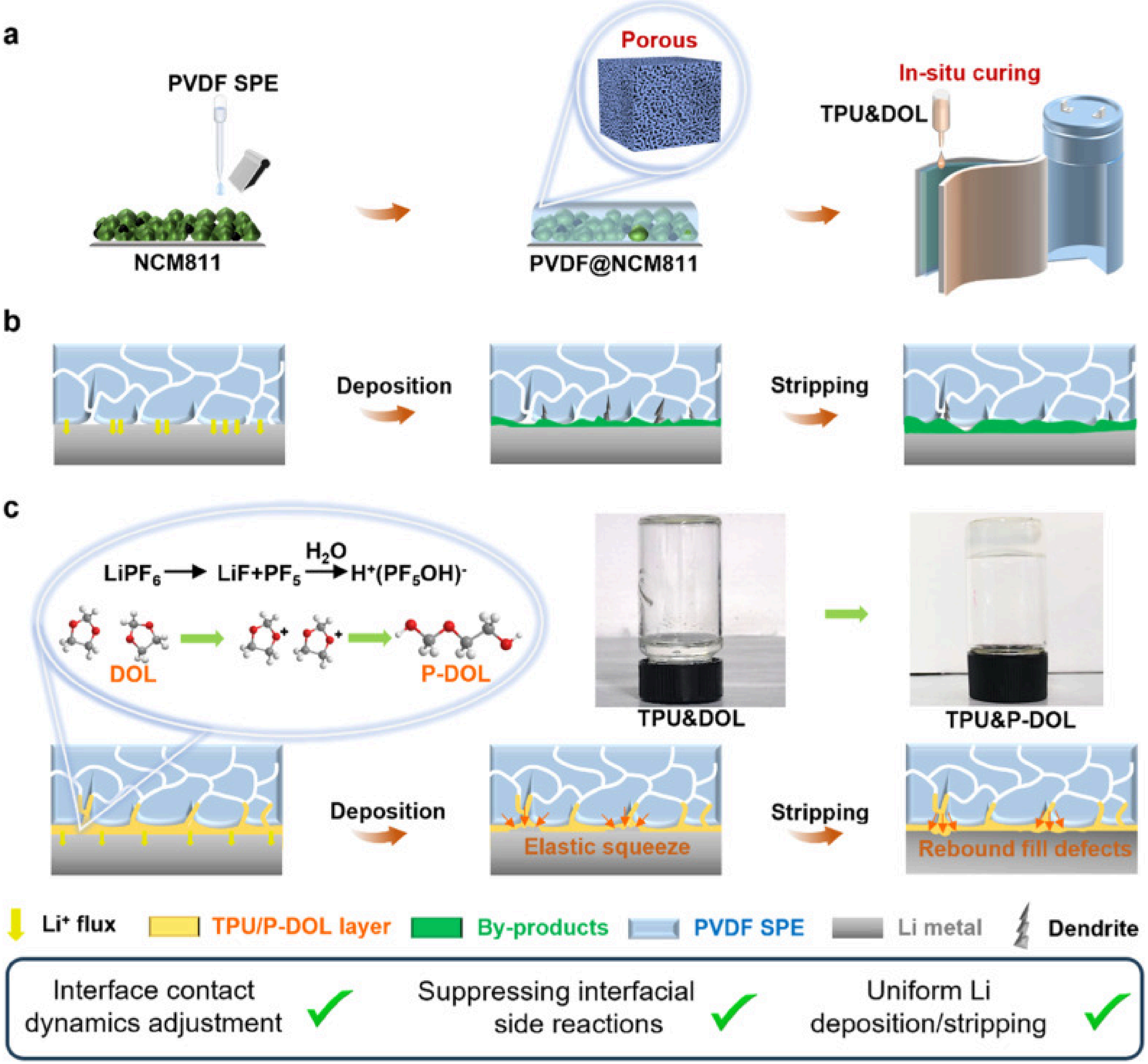
a
PVDF SPE
NCM811
Porous
PVDF@NCM811
In-situ curing
TPU&DOL
b
Deposition
Stripping
c
LiPF6 → LiF+PF5 →H2O H+(PF5OH)-
DOL
P-DOL
TPU&DOL
TPU&P-DOL
Deposition
Elastic squeeze
Stripping
Rebound fill defects
Li+ flux
TPU/P-DOL layer
By-products
PVDF SPE
Li metal
Dendrite
Interface contact dynamics adjustment
Suppressing interfacial side reactions
Uniform Li deposition/stripping

Fig. 1. (a) Preparation process of integrated PVDF@NCM811 and the in-situ cured ultrathin interfacial layer. Schematic illustration of the mechanism for enhancing the electrochemical stability of the PVDF/Li interface and uniformity of Li deposition. (b) Uneven Li deposition due to the porous nature of the PVDF-based SSE. (c) The TPU/P-DOL interface layer inhibits side reactions by avoiding direct contact between PVDF and Li and dynamically regulates the interface contact.

Currently, various methods have been developed to mitigate these issues and thereby enhance contact tightness and chemical stability. Reducing morphological porosity [[12], [13], [14], [15]], modifying the molecular chain structure [16,17], anchoring DMF solvents by fillers [[18], [19], [20]], in-situ formation of stable interfaces by additives [[21], [22], [23], [24]], hybrid/layered structured SSEs [[25], [26], [27]], adjusting solvent content/type [28,29], and thermal-electrochemical treatment [30] have been shown to contribute to an improved cycling stability of LMBs using PVDF as SSE. Most of the abovementioned methods inhibit the continuous occurrence of PVDF reduction and DMF-induced side reactions to a certain extent, but the accumulation of by-products and Li dendrites after long cycling still generates interfacial gaps, thus rendering Li^+ transport and Li deposition less and less uniform.

Although reducing the DMF content will as intended inhibit undesirable side reactions, an excessive reduction is also known to adversely affect the interfacial transport properties [31]. This is not only due to the critical role of DMF in maintaining sufficient ionic conductivity, but also because DMF contributes to the formation of structures that are microscopically similar to those in localized high-concentration electrolytes [32,33] and will initiate the generation of LiF-rich SEI components when the PVDF/DMF system is in contact with lithium metal [34]. Therefore, a simple approach for simultaneously achieving increased morphology density, suppression of side reactions, and dynamic regulation of interfacial adherence without the need for precise restriction of DMF solvent content are particularly important.

Here, we design an ultrathin, highly elastic, and high dielectric constant interfacial layer between PVDF SSE and Li metal anode by triggering in-situ curing of a thermoplastic polyurethane (TPU)-containing 1,3-dioxolane (DOL) precursor solution via Lewis acid groups generated by trace hydrolysis of $LiPF_6$ (Fig. 1). The framework formed by the TPU limits the contact of the reactive liquid solvent such as DMF with the Li, leading to a significant reduction in interfacial side reactions. In addition, the ether-based electrolyte DOL is more compatible with Li metal, with the high dielectric constant of TPU facilitating dissociation of lithium salts and more $TFSI^-$-anion reduction, jointly forming a more stable and robust solid electrolyte interphase (SEI). More importantly, TPU greatly improves the elasticity and adhesion of the interfacial layer, maintaining a tight fit and stable interface impedance by dynamically adjusting the interface contact. With the above benefits, Li||NCM811 full cells retain a specific capacity of 136 mA h g^{-1} after 800 cycles at 1 C. This work provides a practical interface optimization strategy for the development of ultra-long-life solid-state LMBs with dynamically self-regulating interfacial stability.

2. Results and discussion

2.1. Inhibition of side reactions through TPU/P-DOL interfacial layer

It is widely recognized that even over-drying PVDF SSE still cannot completely remove the DMF solvents, instead resulting in decreased flexibility, as well as the degradation of mechanical properties and ionic conductivity [8]. To investigate the content of residual DMF within PVDF, three sets of samples were prepared for thermogravimetric analysis (TGA) varying the drying temperature and duration. As shown in Fig. S1, increasing the drying temperature and time effectively eliminates free DMF (120–160 °C), reducing its content to well below 10 wt. %. In contrast, $[Li(DMF)_x]^+$ species exhibit minimal thermal decomposition even above ~170 °C, indicating strong coordination and high thermal stability. Furthermore, the presence of the TPU/P-DOL interfacial layer does not significantly alter the residual DMF composition for PVDF samples dried under the same conditions.

Therefore, it is particularly crucial to inhibit the side reaction between trace solvent residues and Li metal. To confirm the higher stability between the in-situ cured TPU/P-DOL interfacial layer and Li metal compared to the PVDF SSE, the precursor solution for the interfacial layer and PVDF electrolyte was simultaneously poured onto a Li metal surface to observe the interface evolution with and without TPU/P-DOL layer (Fig. 2a). The bare PVDF polymer darkened after only 2 h in contact with Li and peeled off after 24 h, exposing the obviously side-reacted and darkened Li metal surface. However, the cured precursor solutions TPU/P-DOL and P-DOL reacted minimally with Li metal throughout the whole in-situ curing process, with no significant color change.

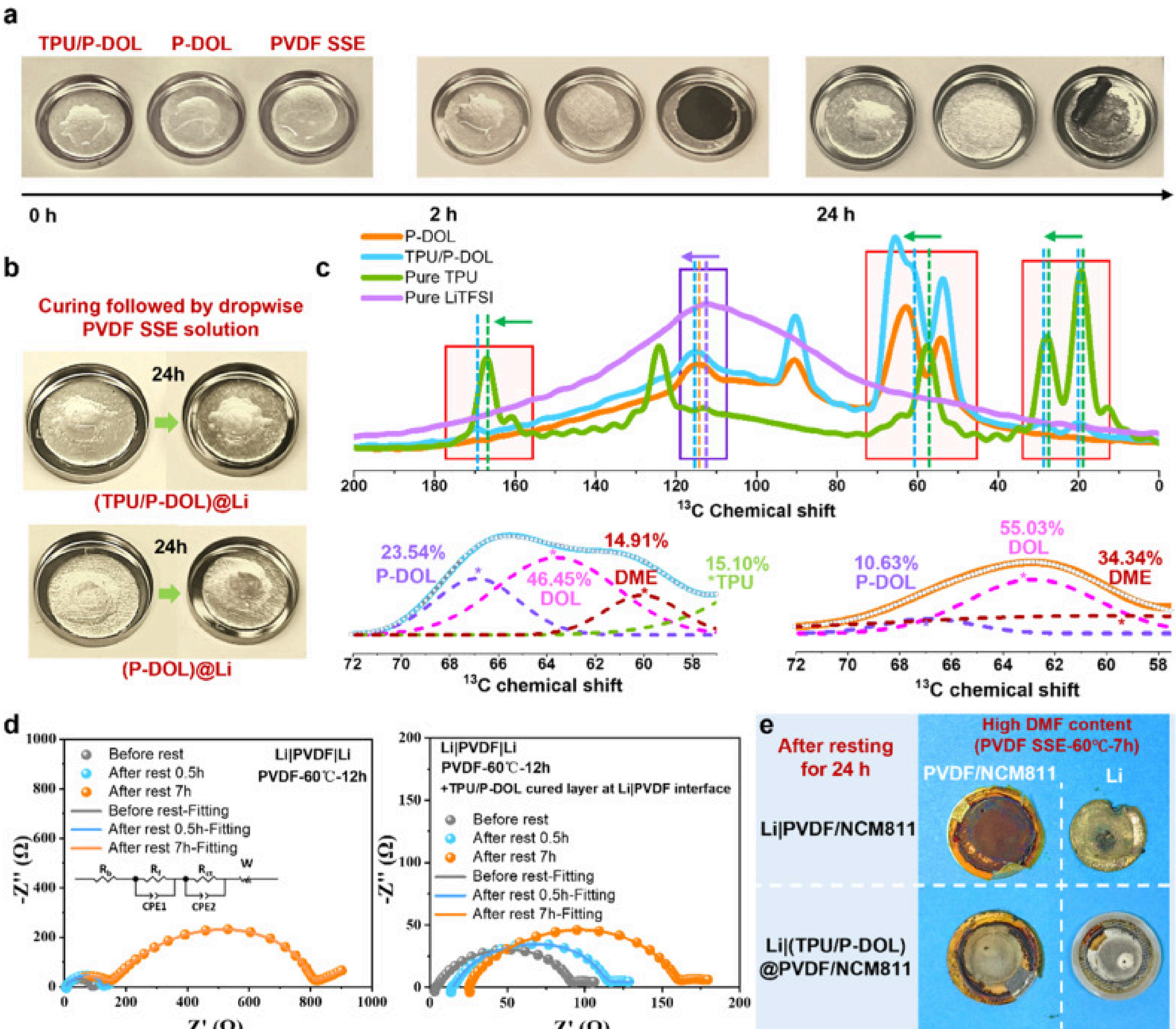

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Fig. 2. (a) Digital images of the gradual evolution of Li foils during the reaction with TPU/P-DOL and P-DOL precursor solution and PVDF SSE solution. (b) Digital images of the gradual evolution of Li foils coated with different cured layer after casting PVDF SSE solution. (c) Solid-state NMR spectroscopy of ^{13}C in pure TPU, P-DOL and TPU/P-DOL cured membrane. (d) EIS impedance spectra of Li||Li symmetric cells after different resting times. (e) Optical images of the interior of Li||NCM811 cells with and without TPU/P-DOL interfacial layers after resting.

To investigate which of the two cured interfacial layers is more resistant to DMF solvent penetration within the PVDF-based SSE, equal volumes of PVDF electrolyte precursor solutions were poured onto Li metal surfaces that had been coated with a cured interfacial layer via spin-coating to observe color changes (Fig. 2b). It is obvious that the curing layer containing TPU is more conducive to blocking DMF penetration and direct contact with Li metal, thus preventing side reactions from occurring (no color change). As can be seen from Figure S2, this is due to the fact that the introduction of TPU into the precursor solution of the interfacial layer is more conducive to its rapid curing into a thin film, which plays a critical role in improving the continuity, uniformity, and completeness of the curing process by providing physical support and structural regulation. Once initiated, the polymerization proceeds efficiently. The ^{13}C ssNMR spectrum in Fig. 2c shows that after the same curing time, the ratio of polymerized DOL (23.54 %) at a chemical shift of 66.9 ppm to unpolymerized DOL (46.45 %) at 63.66 ppm in the TPU-containing cured interface layer is 0.51, which is significantly higher than the ratio of P-DOL (10.63 %) to DOL (55.03 %) of 0.19 in the interface layer without the TPU [35,36]. This further confirms the optical observation that the addition of TPU facilitates the rapid film formation of the precursor solution and explains the underlying reason, which is that TPU facilitates the ring-opening polymerization of DOL and enhances the degree of polymerization.

To verify that the in-situ curing layer can continuously inhibit the side reactions during the actual battery assembly and curing process, symmetric cells were assembled and the variation of impedance over extended cycling was observed (Fig. 2d and Fig. S3). The fitting results show that for the symmetric cell with the TPU/P-DOL interfacial layer, the interfacial charge transfer resistance (R_{ct}) increased from 23.52 Ω to 113.3 Ω after resting for 7 h, which is significantly lower than that (from 6.71 Ω to 638.5 Ω) of the cell without the interfacial layer. The results show that both the TPU/P-DOL and P-DOL interfacial layers significantly improve the cell impedance stability compared to the Li|PVDF|Li cell.

It is worth noting that the insulating property of TPU results in a slight increase in the interfacial impedance of the cell with the addition of the TPU/P-DOL cured layer, but the interfacial film formation is more pronounced. Furthermore, the TPU/P-DOL interfacial layer also can still effectively prevent side reactions at an extremely high residual DMF content within the PVDF SSE (60 °C for only 7 h). Compared with the Li|PVDF|NCM811 cell, the extent of color change of Li and PVDF SSE in the cell with the TPU/P-DOL interfacial layer is significantly reduced after resting for 24 h (Fig. 2e). The trace corrosion still observed in this case may be attributed to the ultrathin nature of the TPU/P-DOL interfacial layer, which cannot fully block the penetration of excessive DMF, leading to slight side reactions. Nevertheless, the interlayer still effectively suppresses most side effects.

2.2. Characterization of the cured TPU/P-DOL interfacial layer

To explore the effects of in-situ casting of PVDF electrolyte precursor solution and in-situ curing of TPU/P-DOL ultrathin layer on the cathode/electrolyte interface and anode/electrolyte interface, the completely cured cells were disassembled and characterized by SEM. As shown in Fig. 3a, in-situ casting of the PVDF electrolyte precursor solution onto the surface of the NCM811 cathode significantly enhanced the tight adhesion of the cathode/electrolyte interface, reducing the formation of voids and other defects. Additionally, due to the internal pressure of the cells, the thickness of the TPU/P-DOL interfacial layer after in-situ polymerization dropped to <5 μm. Besides, Fig. 3b illustrates that the rapid evaporation of DMF solvent and related phase separation during the drying process led to a porous structure in the PVDF electrolyte, causing uneven ion transport flux at the interface [12]. Notably, the PVDF electrolyte surface transformed from porous structure to dense morphology after the introduction of the TPU/P-DOL ultrathin layer, resulting in a noticeable suppression of interfacial deposition/stripping "hot spots" (Fig. 3c).

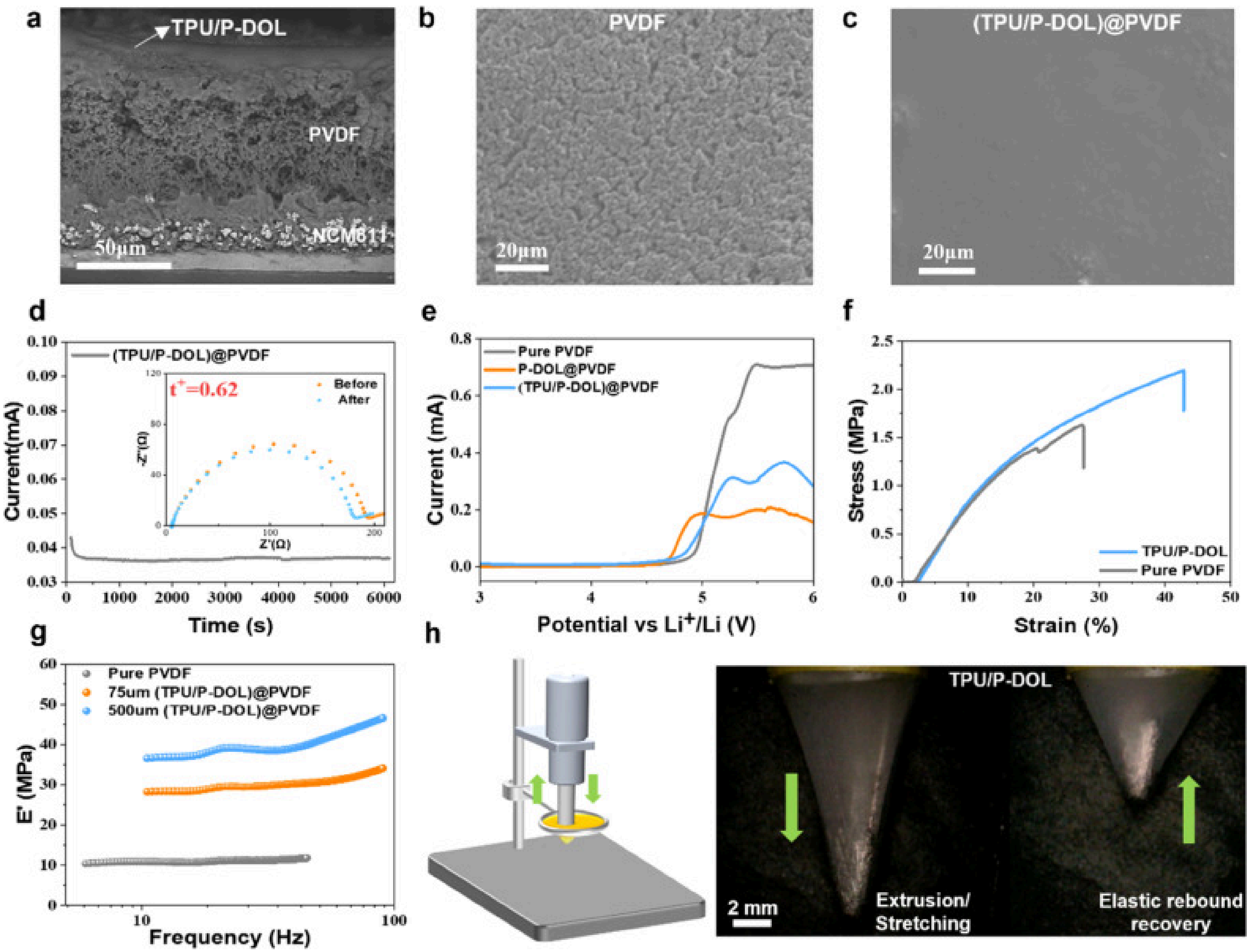

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Fig. 3. (a) SEM images of cross section of cured Li||NCM811 cell after stripping off the Li anode. SEM images of top view of PVDF SSE (b) before and (c) after coated with TPU/P-DOL layer. (d) Ionic transference number test of the (TPU/P-DOL)@PVDF

electrolyte. (e) LSV curves of the PVDF and (TPU/P-DOL)@PVDF electrolytes. (f) Stress-strain test curves of the PVDF and TPU/P-DOL membrane. (g) DMA test of PVDF electrolytes coated with TPU/P-DOL layers of different thicknesses. (h) Instrumentation and optical images for elastic-mechanical testing of TPU/P-DOL membrane.

The ionic conductivity of PVDF SSE at different temperature is shown in Fig. S4, which can reach 2.61×10^{-4} S/cm at 25 °C. The ionic conductivity and activation energy of the PVDF SSE after coating with the TPU/P-DOL interfacial layer were found to be almost unaffected (Fig. S5). In particular, the ionic conductivity decreases only slightly due to the low ionic conductivity of the TPU in the interfacial layer, but the ultrathin thickness of the cured TPU/P-DOL layer ensures it has little effect on the overall conductance of the PVDF-based SSE. Furthermore, the overall ion mobility number of the PVDF electrolyte is clearly elevated after coating with the TPU/P-DOL layer. The doubling of the Li^+ transport number $t^+ = 0.62$ for the interface-modified (TPU/P-DOL)@PVDF (Fig. 3d) when compared to $t^+= 0.30$ for the bare PVDF (Fig. S6) suggests that the $TFSI^-$ anion is strongly anchored in the TPU polymer matrix and dissociated lithium ions now dominate the ion transport. Moreover, the TPU/P-DOL sample exhibits distinct ^{13}C chemical shift changes for both the ester carbon in TPU (red-framed region) and the CF_3 carbon in $TFSI^-$ (purple-framed region) compared to their pure forms (Fig. 2c). Taking the purple-framed region a as an example, a slight ^{13}C chemical shift of LiTFSI is observed in the P-DOL sample compared to its pure form, indicating that the solvation structure or curing process affects the chemical environment of $TFSI^-$. However, the much more pronounced shift in the TPU/P-DOL sample suggests that the variation cannot be solely attributed to solvent or curing effects. Instead, it indicates a cooperative interaction between TPU and $TFSI^-$, likely involving coordination or spatial interactions that alter the local electronic environments, thereby confirming the presence of specific interfacial interactions.

Besides, Li-to-stainless steel cells were assembled for linear sweep voltammetry (LSV) testing (Fig. 3e). The bare PVDF electrolyte showed significant current peaks starting at about 4.5–4.7 V, which can be attributed to the oxidation of the residual organic solvents and the PVDF polymer [31]. In contrast, after introducing the TPU/P-DOL interfacial layer, the oxidation current was significantly reduced, indicating that this interfacial layer suppressed the side reactions and kept the interfacial impedance stable, which worked synergistically with the high ionic mobility number to reduce the polarization and the decomposition of the electrolyte at high voltage.

Fig. 3f displays the comparison of the tensile strength test curves for the PVDF and TPU/P-DOL membrane. The tensile strength of the pure PVDF electrolyte is 1.63 MPa, while the tensile strength of the TPU/P-DOL membrane increases to 2.20 MPa, with the elongation rate increasing from 27.4 % to 42.8 %. Moreover, to quantify the elastic effect of the ultrathin layer, TPU/P-DOL layers were coated onto the surface of the PVDF SSE using a blade with different heights, and dynamic mechanical analysis (DMA) testing was conducted. Fig. 3g shows that the energy storage modulus (E`) of PVDF increases significantly with the rise in the TPU/P-DOL coating thickness, which indicates that PVDF deforms less and less under cyclic stress, and thus TPU/P-DOL has a high elasticity.

To demonstrate the high elasticity of the TPU/P-DOL ultrathin layer more directly and intuitively, it was fixed on an iron ring and then pushed through the ring using a needle to observe the deformation and self-healing ability of the modified electrolyte film as the needle tip moved. Fig. 3h illustrates that the cured TPU/P-DOL membrane is difficult to break even under extreme stretching, and after the stretching force is released, the TPU/P-DOL membrane can automatically return to its original state. However, the PVDF membrane fractured under relatively small deformation, indicating its inferior elasticity and structural stability compared to the TPU/P-DOL cured interfacial layer (Fig. S7).

2.3. Electrochemical performances

To evaluate the effect of the TPU&P-DOL interfacial layer on the electrochemical performance, rate capability tests were first implemented (Fig. 4a). Li||NCM811 full cells were assembled by matching integrated PVDF/NCM811 cathodes. It can be seen that the cell containing a TPU/P-DOL interfacial layer delivers discharge capacities of 172.8, 166.2, 162.3 153.3, 136.9, 128.9, 121.6 and 111.8 mA h g^{-1} at 0.1, 0.3, 0.5, 1, 2, 3, 4 and 5 C, respectively, and the specific capacity remains almost unchanged when the current density is reduced again to 0.1 C. Although the ionic conductivity of the TPU/P-DOL interfacial layer is slightly lower than that of the PVDF electrolyte, it effectively stabilizes the lithium interface by reducing interfacial polarization and enhancing charge transfer kinetics, which synergistically improve the rate performance of the cell. Subsequently, the long-term cycling performance was tested at 1C rate (Fig. 4b). The specific capacity and CE of Li||NCM811 cell with TPU/P-DOL interfacial layer remained at 136 mA h g^{-1} and 99.3 %, respectively, after 800 cycles. However, the specific capacity and CE of the Li||NCM811 cell without interfacial layer decayed rapidly to just 60.9 mA h g^{-1} and 93.8 %, respectively after only 200 cycles, which can be attributed to the severe side reactions increasing the interfacial impedance and reducing the output voltage. To summarize, the performance improvement due to the TPU/P-DOL interfacial layer becomes more and more prominent with increasing rate and cycle number, implying that the interfacial layer achieved stable interfacial impedance, operating voltage plateau and output capacity from the early stage of the cycling through the continuous accumulation of optimization effects such as dynamic adjustment of interfacial contacts, reduction of side reactions and inhibition of Li dendrites.

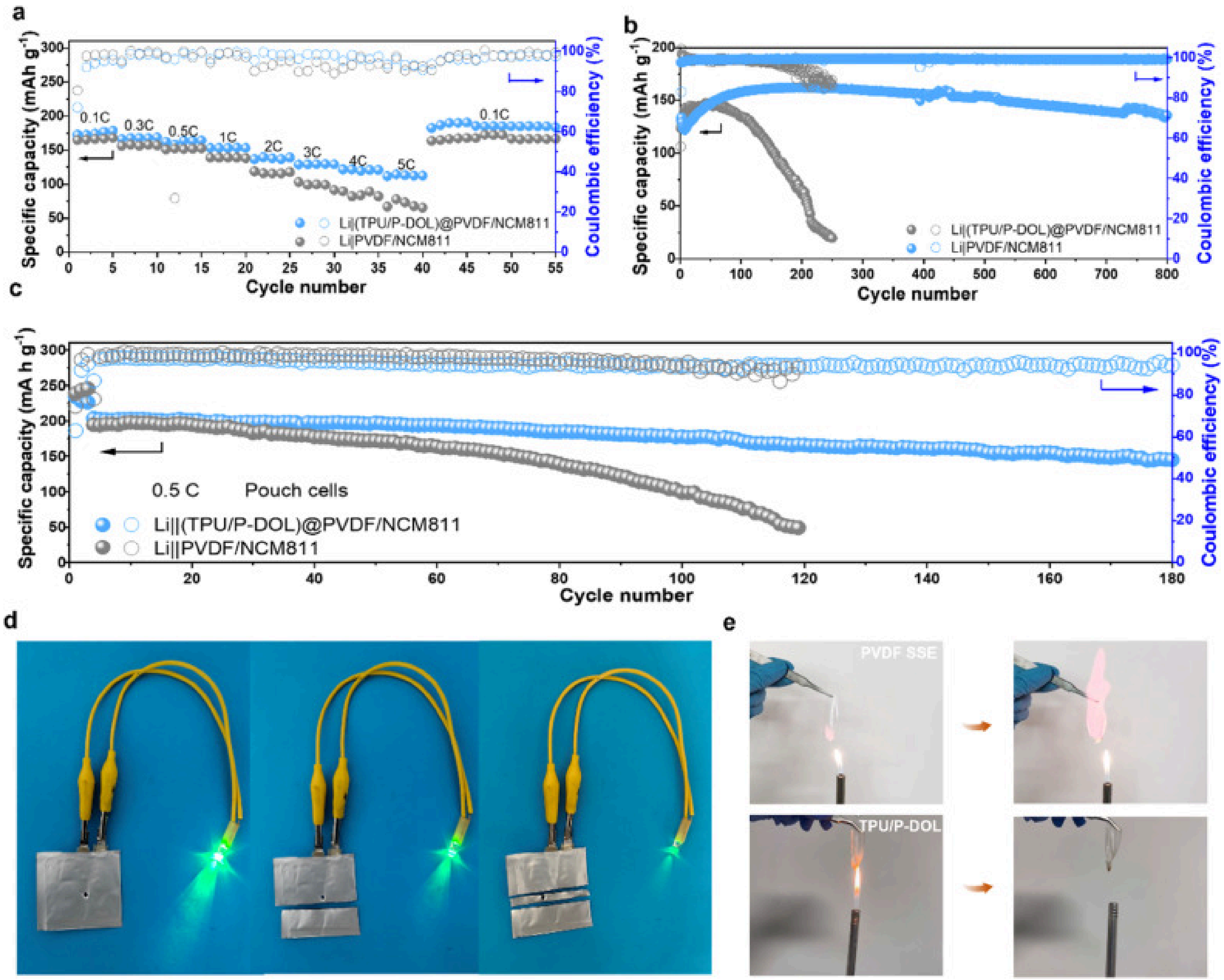

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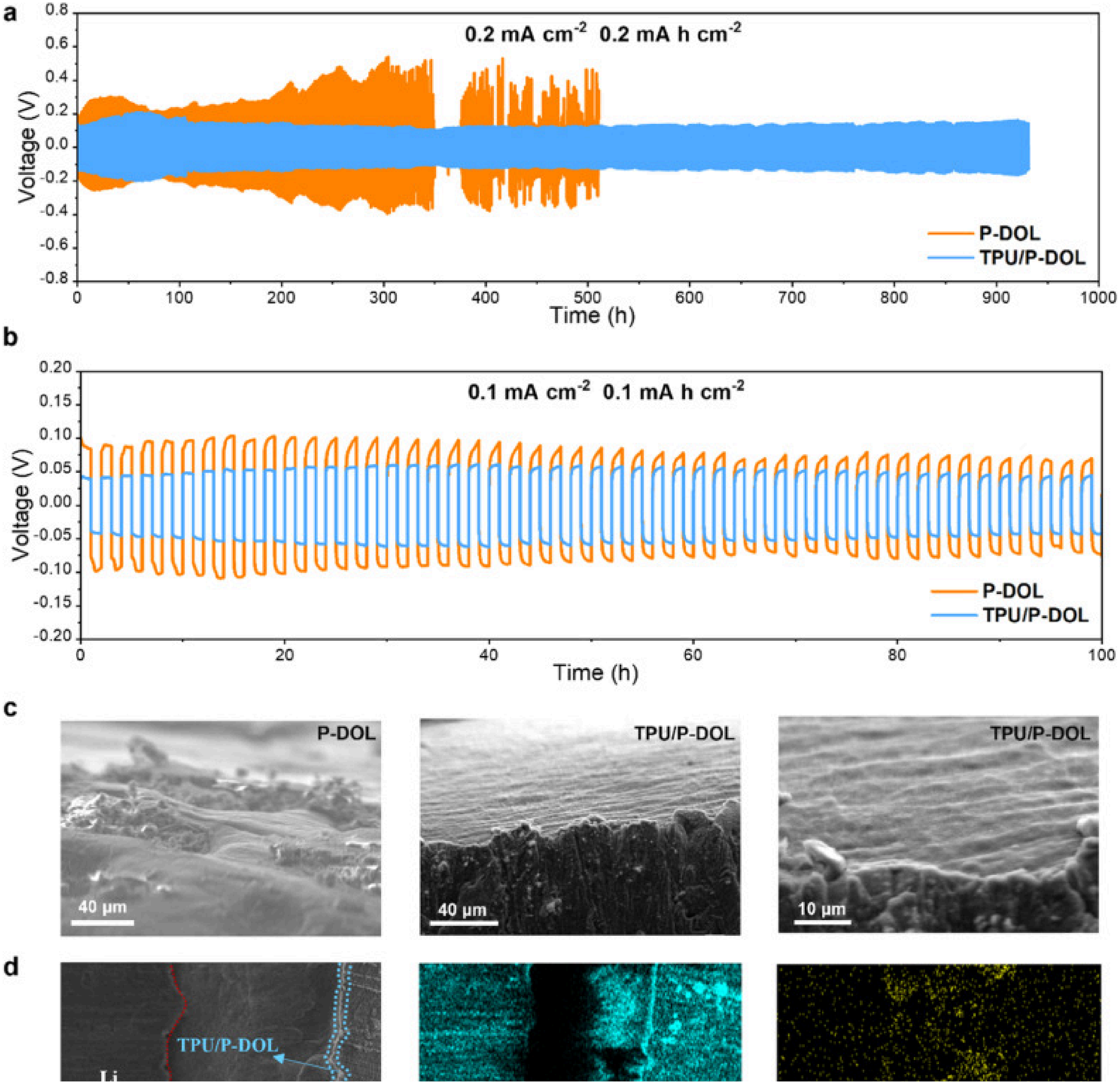
Fig. 4. (a) Rate capacities of Li||NCM811 cells with and without TPU/P-DOL interfacial layer. Long-term cycling performance at 1C of Li||NCM811 in (b) coin cells and (c) pouch cells. (d) Safety test of the pouch cell with TPU/P-DOL interfacial layer. (e) Flammability tests of TPU/P-DOL cured membrane and bare PVDF SSE.

Solid-state pouch cells were then assembled and tested to verify the feasibility of the TPU/P-DOL interfacial layer for mass production and large device applications (Fig. 4c). The specific discharge capacity of the Li|| NCM811 cell without interfacial layer rapidly decays to 98.8 mA h g^{-1} after only 100 cycles, whereas the cell with TPU/P-DOL interfacial layer still maintains the discharge specific capacity of 144.9 mA h g^{-1} after 180 cycles. In addition, under puncture and shear safety tests the pouch cell with the TPU/P-DOL interfacial layer can still operate normally powering LEDs without serious thermal runaway or fire (Fig. 4d). Moreover, the flame retardancy of solid-state electrolytes is also crucial to reduce the risk of fire or explosion in batteries under conditions such as overheating, overcharging or short circuit.

To demonstrate that the flame retardancy of the TPU/P-DOL interfacial layer is significantly stronger than that of PVDF SSE, the TPU/P-DOL membrane and the PVDF SSE were subjected to ignition tests (Fig. 4e). The PVDF-based SSE spontaneously combusted right after contact with the fire, which was caused by both the flammable DMF solvent contained inside and the polymer PVDF substrate itself. In contrast TPU/P-DOL membrane only showed a slight shape shrinkage even after contact with open flames, presumably due to the flame retardant contained in the TPU and the urethane groups (Fig. S8) in its molecular chain decomposing during the heating process and generating intermediates that further inhibit combustion [37].

2.4. Morphology characterizations of Li deposition after cycling

To explore the effect of the TPU in the cured layer on the inhibition of dendrite growth at the interface, symmetric cells were assembled using two types of cured precursor solutions (TPU/P-DOL and P-DOL), and their long cycle performance was tested at a current density of 0.2 mA cm^{-2} and a deposition capacity of 0.2 mA h cm^{-2} (Fig. 5a). The addition of TPU within the curing interfacial layer significantly improved the overpotential stability of the cell during cycling, whereas symmetrical cells with curing layers not containing TPU showed inferior cycle life, with rapid increases in polarization voltage and micro-short circuits occurring after <100 h.


a
0.2 mA cm-2 0.2 mA h cm-2
Voltage (V)
Time (h)
P-DOL
TPU/P-DOL
b
0.1 mA cm-2 0.1 mA h cm-2
Voltage (V)
Time (h)
P-DOL
TPU/P-DOL
c
P-DOL
TPU/P-DOL
TPU/P-DOL
40 μm
40 μm
10 μm
d
TPU/P-DOL
Li

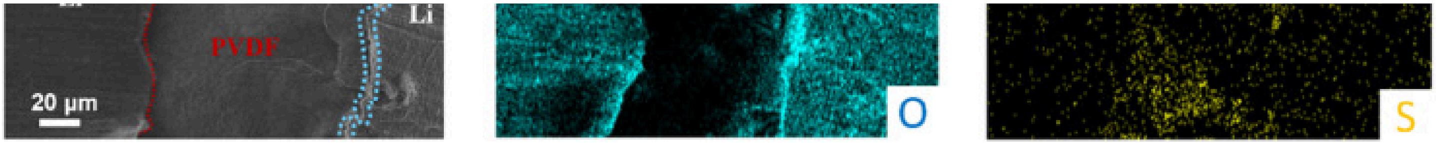


Fig. 5. Voltage profiles of symmetrical cells with different interfacial layer cycled at (a) 0.2 mA cm^{-2}, 0.2 mA h cm^{-2} and (b) 0.1 mA cm^{-2}, 0.1 mA h cm^{-2}. (c) SEM images of the cycled Li surface from symmetric cells after 100 cycles at 0.1 mA cm^{-2} and 0.1 mA h cm^{-2}. (d) SEM images and EDS mapping of the interfaces between the Li metal electrode and PVDF electrolytes after cycling.

To investigate the underlying cause of the enhanced stability, the cell was disassembled after cycling for 100 h at a current density of 0.1 mA cm^{-2} and a deposition capacity of 0.1 mA h cm^{-2} (Fig. 5b) and the morphology evolution at the interface was observed. It can be clearly seen from Fig. 5c that the surface of the Li electrodes matched with the cured layer containing TPU is much flatter and denser, while the surface of the Li electrodes matched with the cured layer not containing TPU shows obvious pits and dendritic morphology.

The cross-sectional energy dispersive spectrometer (EDS) images of symmetric cells were also scanned to explicitly inspect the TPU/P-DOL cured layer after cycling. Here, the TPU/P-DOL interfacial layer is inherently rich in oxygen-containing species due to its chemical composition. Specifically, TPU contains abundant C=O and C–O groups within its polymer backbone. DOL, as a cyclic ether, undergoes ring-opening polymerization to form polyether chains with multiple oxygen atoms, while DME introduces additional linear ether units. To further confirm the composition and thickness of this interlayer, EDS elemental mapping was conducted (Fig. 5d). The results confirm the presence of a uniformly distributed, oxygen-rich interfacial layer with a thickness of <10 μm. In addition, higher-magnification SEM and EDS images (Fig. S9) of the symmetric cell after 25 cycles further reveal that the interfacial layer maintains its structural integrity, with a consistent thickness of approximately 10 μm.

To further demonstrate the high electrochemical stability of the TPU/P-DOL interfacial layer with Li metal, and its high elasticity-induced redistribution of interfacial stresses in liquid cells can inhibit the growth of dendrites to form a dense deposition morphology, the TPU/P-DOL interfacial layer was cured and attached to one side of the polyethylene separator and assembled into a liquid Li||Cu half cells. Then, the cell deposited 5 mA h cm^{-2} of Li metal at a current density of 1 mA cm^{-2} and the Li deposition morphology on the surface of the Cu foil was observed by atomic force microscopy (AFM) and SEM (Fig. S10). The images reveal that after the introduction of the TPU/P-DOL interfacial layer, the Li deposition on the surface of the Cu foil exhibits an island-like bulk morphology. During half-cell discharge, Li^+ is released from the Li negative electrode and deposited on the Cu collector (positive electrode). During charging, the source of Li returning to the negative electrode is entirely derived from Li deposition on the surface of the Cu foil during the first discharge, and thus this dense deposition morphology on the surface of the Li-free Cu foil better accounts for the electrochemical stability of the cured interfacial layer to Li metal.

2.5. Optimization mechanism of TPU/P-DOL interfacial layer

After placing the TPU polymer particles into the commercial carbonate-based electrolyte for a certain period of time, it can be clearly seen that the electrolyte is absorbed into the interior of the TPU polymer particles (Fig. S11). Although the TPU polymer still maintains the particle morphology, an obvious swelling phenomenon has occurred. Therefore, the TPU polymer can adsorb and confine the DME and unpolymerized DOL solvent in the cured interfacial layer, which effectively reduces the mobility of the cured precursor solvent inside the cell and the undesired consumption.

The highly elastic ultrathin TPU/P-DOL modified layer between the Li metal anode and the PVDF SSE interface shown in Fig. 5c is able to buffer and weaken the mechanical stress caused by the volume change of the deposited Li metal. In addition, the highly elastic properties of the interfacial layer allow it to dynamically maintain a tight fit, boosting the stability and integrity of the interface during cycling, avoiding the occurrence of cracks and interfacial separation. To illustrate the function of a high elastic modulus of the interfacial layer more intuitively, COMSOL simulations of the interfacial pressure distribution have been implemented. Simulation results in Fig. 6a indicate that the characteristic Li dendrite morphology with high aspect ratio is particularly susceptible to the influence of TPU/P-DOL modified layer, the pressure on the tip of these dendrites decreases from 0.02262 N m^{-2} (I) to 0.02184 N m^{-2} (II) and 0.02107 N m^{-2} (III) when the Young's modulus of the interfacial modified layer rises from 1×10^7 Pa to 1×10^8 Pa and 2×10^8 Pa. Thus, an interfacial layer with strong flexibility and elasticity helps to slow down the growth rate of lithium dendrite tips, which ultimately promotes a dense deposition morphology.

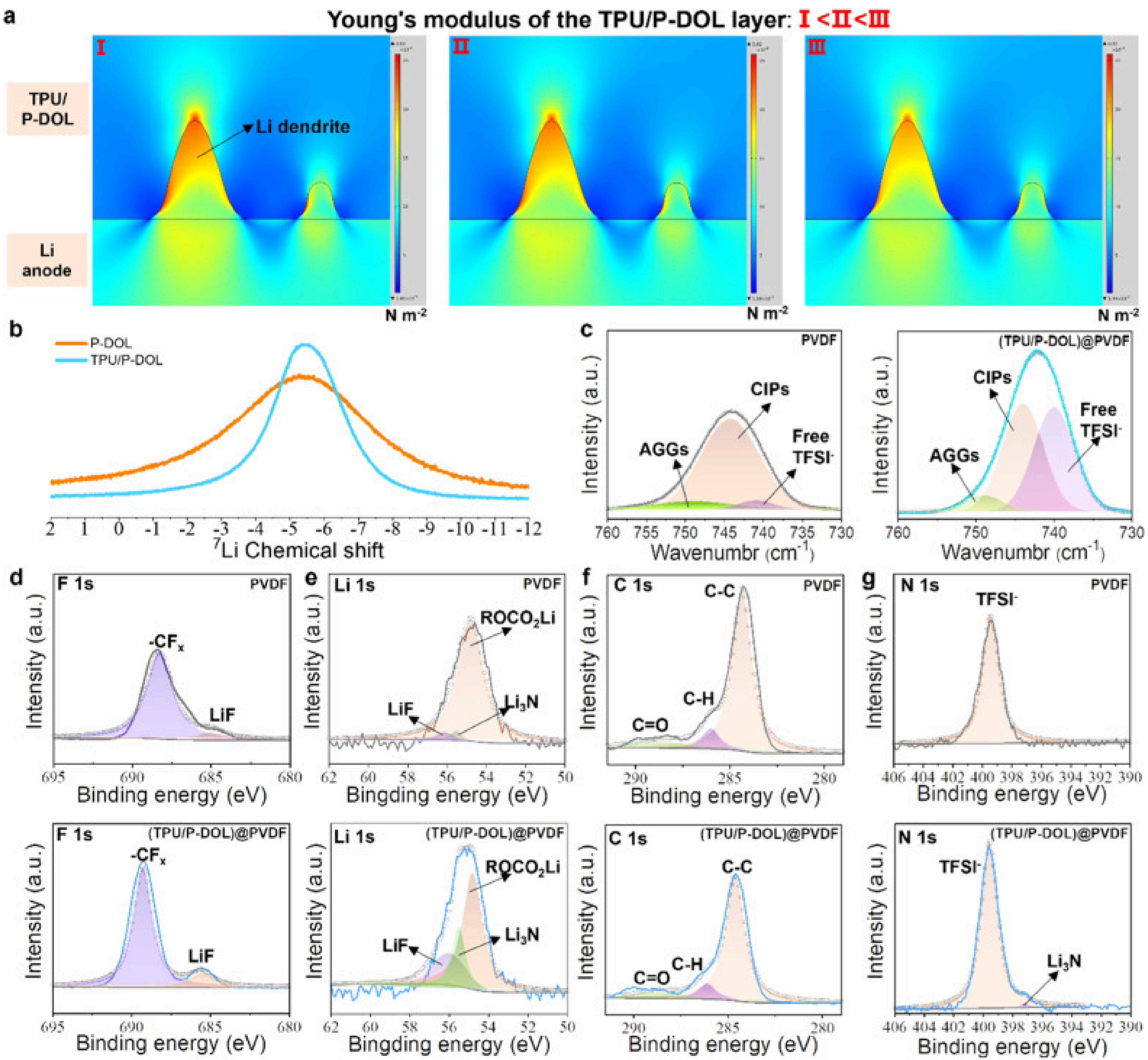

a
Young's modulus of the TPU/P-DOL layer: I < II < III
TPU/ P-DOL
Li anode
Li dendrite
N m-2
b
P-DOL
TPU/P-DOL
7Li Chemical shift
c
PVDF
(TPU/P-DOL)@PVDF
AGGs
CIPs
Free TFSI-
Intensity (a.u.)
Wavenumbr (cm-1)
d F 1s
-CFx
LiF
e Li 1s
ROCO2Li
Li3N
Bingding energy (eV)
f C 1s
C-C
C-H
C=O
g N 1s
TFSI-
Binding energy (eV)

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Fig. 6. (a) COMSOL simulation image of the pressure distribution at the Li/(TPU/P-DOL) interface. (b) ^{7}Li SSNMR Spectroscopy of the P&M and TPU/P-DOL cured layer. (c) Raman spectra of the PVDF SSE and TPU/P-DOL cured layer. XPS spectra of (d) F 1 s, (e) Li 1 s, (f) C 1 s and (g) N 1 s collected from the cycled Li metal anode surface with and without the TPU/P-DOL cured layer.

Moreover, the added TPU inside the cured layer can also act synergistically to form a more stable Li/PVDF interface by chemical interaction. The solid-state nuclear magnetic resonance (ssNMR) spectroscopy of the cured samples reveals that the TPU/P-DOL membrane has higher ^{7}Li chemical shift peaks and narrower peak widths compared to the P-DOL membrane (Fig. 6b). This suggests that the added TPU increases the dissociation of lithium salts enhancing the homogeneity of the chemical environment around the Li^{+}.

Therefore, the coordination state between $TFSI^{-}$ and Li^{+} in PVDF SSE and TPU/P-DOL cured membrane was tested through deconvoluting and comparing the Raman spectroscopy to verify the above hypothesis. The Raman bands at 720 cm^{-1}, 732 cm^{-1}, and 746 cm^{-1} in Fig. 6c refer to free $TFSI^{-}$ anions, contact-ion pairs (CIPs, one $TFSI^{-}$ anion interacting with one Li^{+}), and aggregates (AGGs, one $TFSI^{-}$ anion interacting with two or more Li^{+}), respectively [38]. TPU/P-DOL interfacial layer has the highest percentage of free $TFSI^{-}$, indicating that the interaction between Li^{+} and $TFSI^{-}$ is weakened, which allows more free $TFSI^{-}$ anions to be released in the cured layer and as presumed improves the dissociation of lithium salts [25].

The chemical compositions of the Li/PVDF interface layers in Li||NCM811 cells after cycling at 1C (Fig. S12) are analyzed by X-ray photoelectron spectroscopy (XPS). In the F 1 s spectra (Fig. 6d), the peak intensity of LiF is obviously greater when applying TPU/P-DOL cured layer, which is mainly ascribed to the decomposition of $TFSI^{-}$ at the Li/PVDF interface [13]. Similarly, it can be seen that peaks at 55.5 eV and 56.0 eV in the Li 1 s spectrum (Fig. 6e), corresponding to Li_3N and LiF, respectively, are also much more pronounced for the sample matched with the TPU/P-DOL cured layer [22]. This is consistent with previous findings that moderate $TFSI^{-}$ dissociation, when coupled with matrix confinement, facilitates the formation of LiF-rich and uniform SEI layers, contributing to enhanced interfacial stability [39]. In contrast, the peak intensities referring to C—C (284.8 eV)/C—H (286.1 eV) and C=O (288.8 eV) in the C 1 s spectra (Fig. 6f) are markedly lowered in the sample with the TPU/P-DOL cured layer, indicating that the undesirable decomposition of PVDF polymer and oxidation of N—C=O in DMF are both suppressed by the TPU/P-DOL cured layer [11]. Notably, the presence of distinct CF_x signals in the F 1 s spectrum, contrasted with ambiguous CF_x signals in the C

1 s spectrum, may be attributed to changes in the chemical environment, peak overlap, or relatively weak signal intensity, which has also been reported in other studies of PVDF-based systems. In the N 1 s spectra (Fig. 6g), the peaks at approximately 399.4 eV and 397.0 eV are assigned to the signals for LiTFSI fragments and Li_3N, respectively, and the latter signal is enhanced for the sample with the TPU/P-DOL cured layer. Based on the above results, the TPU/P-DOL ultrathin layer favors the formation of a LiF/Li_3N-rich SEI, which has superior Li^+ conductivity and excellent electrochemical stability against Li metal [25]. Such inorganic-rich SEI is known to suppress vertical Li dendrite growth but facilitate planar Li deposition [40].

3. Conclusions

In conclusion, the in situ curing of a highly elastic ultrathin layer at the PVDF/Li interface is demonstrated as a facile, scalable approach to address interfacial issues in PVDF-based solid-state batteries. The incorporation of TPU significantly enhances the elastic modulus of the interfacial layer, enabling it to absorb and buffer the local mechanical stresses caused by the uneven Li deposition and the massive volume change of the Li anode during cycling. In addition, the TPU/P-DOL ultrathin interlayer realizes dynamic redistribution of stress and ensures intimate contact at the Li/PVDF interface, effectively suppressing the Li dendrite growth and the interfacial voids formation. These features contribute to uniform ion transport across the interface and long-term interfacial stability. As a result, the symmetric Li|PVDF|Li cell using the TPU/P-DOL interlayer operates over 900 h with stable overpotential, in stark contrast to the symmetrical cell using TPU-free interlayer (P-DOL), which only cycles for <350 h at 0.2 mA h cm^{-2} with severe overpotential fluctuation. For the Li|PVDF|NCM811 full cell assembling with the TPU/P-DOL interlayer, the specific capacity of 136 mA h·g^{-1} and the coulombic efficiency of 99.27 % were obtained at the 800th discharge cycle under 1 C. This work provides a scalable interfacial engineering strategy to promote the practical deployment of PVDF-based polymer electrolytes in solid-state lithium metal batteries, particularly for high-capacity applications such as electric vehicles and grid-level energy storage.

4. Experimental section

4.1. Materials

LiTFSI, $LiPF_6$, 1,2-Dimethoxyethane (DME), DOL, and DMF were procured from Merck Co., Ltd. NCM811 powder, super P and PVDF were purchased from Guangdong Canrd New Energy Technology Co., Ltd. in China. The NCM811 is of single crystalline microstructure. TPU was purchased from Shandong Yousuo Chemical Technology Co. in China.

4.2. Preparation of in situ cured precursor solutions

The preparation of the TPU-containing DOL precursor solution involved the following steps: The TPU (~3–5 wt. %) was completely dissolved in DOL/DME (molar ratio 1:1) solvent under constant stirring at a temperature of 60 °C to obtain a clear polymer solution. Afterwards, 2 M $LiPF_6$ and 1 M LiTFSI were added in batches, and stirred quickly at room temperature until a transparent solution is formed before curing.

4.3. Electrode/electrolyte fabrication and cell assembly

PVDF was first dissolved in DMF to form a 5 wt. % solution to ensure homogeneous dispersion of all components. Then, the cathode was prepared by dispersing NCM811, PVDF, super P and LiTFSI with a weight ratio of 70:10:13:7 in DMF solvent. After stirring uniformly, the slurry was bladed onto carbon-coated aluminum foil and dried under vacuum at 80 °C for 24 h. The areal loading of the active NCM811 cathode material in the full cells is 2.0 mg/cm^2.

PVDF powder and LiTFSI were dissolved in DMF solvent at a mass ratio of 1:2 and stirred until a transparent solution was obtained. The resulting electrolyte precursor solution was then cast directly onto a glass plate or the cathode surface using a doctor blade and vacuum-dried at 60 °C for 24 h. Finally, either a freestanding PVDF solid-state electrolyte or an integrated PVDF/NCM811 was prepared.

For the assembly of coin cells, ~10 μL precursor solution was added between PVDF SSE and Li metal anode. Then the cells were assembled and rested for 6–12 h until fully in-situ cured before testing (Fig. 1a). For pouch cells test, the NCM811 cathode area is 3 × 3 cm and the ultra-thin Li metal anode size is 3.5 × 3.5 cm (20 μm Li, ~ 4 mA h cm^{-2}). All cells were assembled inside a high-purity Ar-filled glove box.

4.4. Materials characterization

The morphologies and energy dispersive spectrometry (EDS) of the samples were characterized by the Phenom ProX scanning electron microscope and Gemini 500 SEM. X-ray photoelectron spectroscopy (XPS) of sample surfaces were measured by Thermo Fisher ESCALAB Xi+ instrument. Raman spectra were obtained with a Raman spectrometer (RENISHAW, Invia Reflex). The chemical environments of ^{13}C and ^{7}Li were examined by solid-state nuclear magnetic resonance spectroscopy (Bruker 400WB). The morphologies of the deposited Li were conducted by a commercial AFM system (MFP-3D, Asylum Research, Oxford Instruments, CA, USA). Dynamic mechanical analysis (DMA) was performed using a Netzsch DMA 242E instrument under the following conditions: frequency range of 0.1–100 Hz, static force of 0.2 N, dynamic force of 0.05 N, and isothermal testing at 25 °C.

4.5. Electrochemical tests

The full cell was assembled by matching the Li anode with the integrated PVDF SSE/cathode (PVDF/NCM811), and ~10 μL of TPU/DOL curing precursor solution was added dropwise between the anode/electrolyte interface before sealing. For investigating the effect of the TPU/DOL addition on the interface evolution, Li||NCM811 cells were disassembled after resting for 5 h, and it was found that the anode/electrolyte interface was free of visible liquid and the side reactions were significantly suppressed (Fig. S13). All the Li||NCM811 coin cells were tested between 3.0–4.3 V by Land Battery Test System (Wuhan, China) and Neware Battery Test System at room temperature (25 °C). Electrochemical impedance spectroscopy (EIS) measurements were performed using a Solartron 1260 electrochemical workstation in the frequency range from 1 MHz to 100 mHz with a perturbation voltage of 10 mV. To test the ionic conductivity of PVDF SSE, the free-standing PVDF SSE membrane was obtained by vacuum drying at 60 °C for 24 h after doctor blading on a glass plate. For pouch cells test, the NCM811 cathode area is 3 × 3 cm and the ultrathin Li metal anode size is 3.5 × 3.5 cm (20 μm Li, ~ 4 mA h cm^{-2}).

4.6. COMSOL simulation of stress distribution

The surface stress distribution was solved by a finite element method on the COMSOL Multiphysics 5.6 platform with an adaptive grid. The Li metal anode was modelled as a 200 μm × 30 μm × 100 μm metal plate, and the cross section was selected to visualize the stress distribution. A number of bumps (0–3 μm) were randomly distributed on the metal plate to simulate an uneven anode surface, so as to observe the effect of the added elastic layer on the stresses around the bumps. The thickness of the elastic interface layer was designed to be 6 μm, and the bottom end of the Li anode was used as a fixed constraint (unable to deform). A force of F = 0.01 N m^{-2} was uniformly applied perpendicular to the surface of the elastic layer at the top end of the elastic layer, and the density of the lithium metal was set to be 530 kg m^{-3}, Young's modulus to be 4.9 GPa, and Poisson's ratio to be 0.36 [[41], [42], [43], [44]]. The density of the elastic layer was set to be 1250 kg m^{-3}, with the Young's modulus designed in three types of 10 MPa, 100 MPa, and 200 MPa, and the Poisson's ratio was set to 0.4 [45]. Steady state simulations were used to observe the magnitude of the stresses on the lithium metal bumps, and all the simulations were carried out at room temperature (298 K) and ambient pressure (101.3 kPa).

CRediT authorship contribution statement

Kaiming Wang: Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Jingjin Xu:** Investigation, Data curation. **Ming Xu:** Visualization, Methodology. **Fei Shen:** Methodology, Conceptualization. **Liang Zhang:** Investigation, Data curation. **Zhiyi Zhou:** Visualization, Software, Investigation. **Aaron Jue Kang Tieu:** Writing – review & editing, Visualization. **Haowen Wu:** Writing – review & editing, Validation. **Xiaogang Han:** Supervision, Funding acquisition. **Stefan Adams:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is supported by State Key Laboratory of Advanced Electromagnetic Technology (Grant No. AET 2025KF004), the National Key R&D Program of China (Grant No 2020YFA0710500), Singapore Ministry of Education (Grant No T2EP50122–0021) and National Research Foundation, Singapore (Grant No NRF-CRP30–2023–0001). The authors acknowledge the basic scientific research business expenses Program of Xi’an Jiaotong University supported by “the Fundamental Research Funds for the Central Universities” (Grant No xzy022022053), the Independent Research Project of the State Key Laboratory of Electrical Insulation and Power Equipment (Grant No EIPE23303) and the China Sholarship Council for financial support. The authors also thank the China Scholarship Council for financial support and the Instrument Analysis Center of Xi'an Jiaotong University and the State Key Laboratory of Electrical Insulation and Power Equipment of Xi'an Jiaotong University for their assistance in characterization.

Appendix. Supplementary materials

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Recommended articles

Data availability

Data will be made available on request.

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